ECEN 743: Reinforcement Learning

Natural Policy Gradient

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References

- [SB, Chapter 13]
- [MMT, Chapter 12]
- [AJKS, Section 3]

Reinforcement Learning Questions

- How do we learn the value of a policy π?
- How do we learn the optimal action-value function Q^*?
- How do we learn the optimal policy π^*?

... without the knowledge of the model P

- Need to learn from the observed sequence of states, actions, and rewards

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$$\Delta(\theta) = \sqrt{\frac{\epsilon}{(\nabla V)^\top G^{-1}(\nabla V)}} \; G^{-1}(\nabla V)$$

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- Using $\lambda^{\star}$, we get $\delta^{*} = \delta(\lambda^{\star})$ as

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- Is (∇V) the best direction?

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- We will use Kullback–Leibler divergence as a measure of distance between two policies

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- Define the expected KL divergence between two polices π_1 and π_2 as

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Taylor Series Expansion of KL Divergence

- Taylor series expansion of a function $f(y)$ around y_0

$$f(y) \approx f(y_0) + (y - y_0)^\top \nabla_y f(y)|_{y_0} + (y - y_0)^\top \nabla_y^2 f(y)|_{y_0} (y - y_0) + \cdots$$

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&+ (\theta - \theta_0)^\top \nabla_\theta^2 D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta)|_{\theta_0} (\theta - \theta_0) + \cdots
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\theta^2 D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta)|_{\theta_0} &= \nabla_\theta^2 \mathbb{E}_{s \sim \rho_{\pi_{\theta_0}}} \mathbb{E}_{a \sim \pi_{\theta_0}(s,\cdot)} \left[\log \frac{\pi_{\theta_0}(s,a)}{\pi_\theta(s,a)} \right] |_{\theta_0} \\
&= -\mathbb{E}_{s \sim \rho_{\pi_{\theta_0}}} \mathbb{E}_{a \sim \pi_{\theta_0}(s,\cdot)} \nabla \left[\frac{\nabla \pi_\theta(s,a)}{\pi_\theta(s,a)} \right] |_{\theta_0} \\
&= -\mathbb{E}_{s \sim \rho_{\pi_{\theta_0}}} \mathbb{E}_{a \sim \pi_{\theta_0}(s,\cdot)} \left[\frac{\nabla^2 \pi_\theta(s,a)}{\pi_\theta(s,a)} - \frac{\nabla \pi_\theta(s,a)\ \nabla \pi_\theta(s,a)^\top}{(\pi_\theta(s,a))^2} \right] |_{\theta_0} \\
&= \mathbb{E}_{s \sim \rho_{\pi_{\theta_0}}} \mathbb{E}_{a \sim \pi_{\theta_0}(s,\cdot)} \left[\frac{\nabla \pi_\theta(s,a)\ \nabla \pi_\theta(s,a)^\top}{(\pi_\theta(s,a))^2} \right] |_{\theta_0}
\end{aligned}
$$

Taylor Series Expansion of KL Divergence

- Taylor series expansion of $D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta)$

$$D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta) = D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_{\theta_0}) + (\theta - \theta_0)^\top \nabla_\theta D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta)|_{\theta_0}$$
$$+ (\theta - \theta_0)^\top \nabla_\theta^2 D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta)|_{\theta_0}(\theta - \theta_0) + \cdots$$

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$$= \mathbb{E}_{s\sim\rho_{\pi_{\theta_0}}} \mathbb{E}_{a\sim\pi_{\theta_0}(s,\cdot)} \left[\frac{\nabla \pi_\theta(s,a)\ \nabla \pi_\theta(s,a)^\top}{(\pi_\theta(s,a))^2}\right]|_{\theta_0}$$
$$= \mathbb{E}_{s\sim\rho_{\pi_{\theta_0}}} \mathbb{E}_{a\sim\pi_{\theta_0}(s,\cdot)} \left[\nabla \log \pi_\theta(s,a)\ \nabla \log \pi_\theta(s,a)^\top\right]|_{\theta_0}$$

Taylor Series Expansion of KL Divergence

- Taylor series expansion of $D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta)$

$$\begin{aligned} D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta) = D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_{\theta_0}) + (\theta - \theta_0)^\top \nabla_\theta D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta)|_{\theta_0} \\ + (\theta - \theta_0)^\top \nabla_\theta^2 D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta)|_{\theta_0}(\theta - \theta_0) + \cdots \end{aligned}$$

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- Fisher information matrix

$$F(\theta_0) = \mathbb{E}_{x \sim \rho_{\pi_{\theta_0}}} \mathbb{E}_{a \sim \pi_{\theta_0}(x,\cdot)} \left[\nabla \log \pi_\theta(x,a)\ \nabla \log \pi_\theta(x,a)^\top \right] |_{\theta_0}$$

Taylor Series Expansion of KL Divergence

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- $\nabla_\theta^2 D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta)|_{\theta_0} = F(\theta_0)$

Natural Policy Gradient

- Using Taylor series expansion of $D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta)$

$$D_{KL}^{\pi_{\theta_0}}(\pi_{\theta_0}, \pi_\theta) \approx (\theta - \theta_0)^\top F(\theta_0)(\theta - \theta_0)$$

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- We we want to find a direction $\delta\theta$, where

$$\arg\max_{\delta} \ \delta^\top(\nabla V_{\pi_\theta}), \quad \text{s.t. } D_{KL}^{\pi_\theta}(\pi_\theta, \pi_{(\theta+\delta\theta)}) \leq \epsilon$$

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- Natural Policy Gradient

$$\theta \leftarrow \theta + \delta\theta$$

$$\delta\theta = \sqrt{\frac{\epsilon}{(\nabla V)^\top F(\theta)^{-1}(\nabla V)}} \ F(\theta)^\dagger (\nabla V)$$

Example: Policy Gradient for LQG

- Consider a scalar Linear Quadratic Gaussian (LQG) problem

$$x_{t+1} = Ax_t + Bu_t + w_t,\ w_t \sim \mathcal{N}(0, \sigma^2),\ R_t = -Qx_t^2 - Pu_t^2$$

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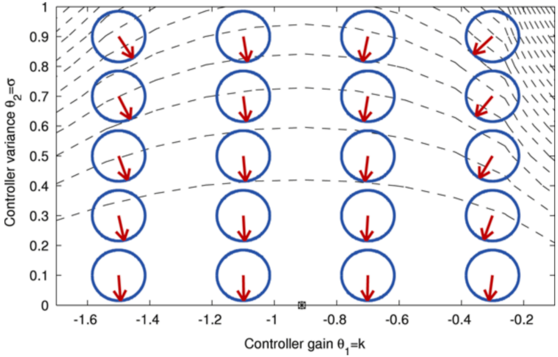
- Perform vanilla PG: $\theta \leftarrow \theta + \alpha \widehat{\nabla V_{\pi_\theta}}$
- How will the vanilla PG perform?

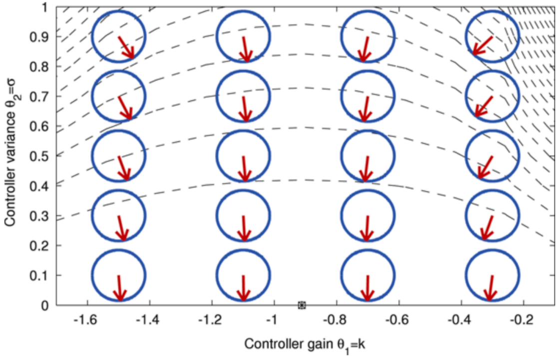
Example: Vanilla Policy Gradient for LQG



- Control policy: $u_t \sim \mathcal{N}(\theta_1 x_t, \theta_2)$, $\theta \leftarrow \theta + \alpha \widehat{\nabla V_{\pi_\theta}}$

Figure from Peters, Jan, and Stefan Schaal. "Reinforcement learning of motor skills with policy gradients", *Neural networks*, 2008

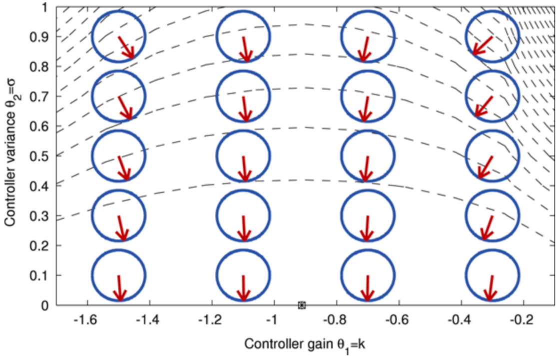
Example: Vanilla Policy Gradient for LQG



- Control policy: $u_t \sim \mathcal{N}(\theta_1 x_t, \theta_2)$, $\theta \leftarrow \theta + \alpha \widehat{\nabla V_{\pi_\theta}}$
- Decrease of exploration has a stronger immediate effect on the expected return. So, the gradient mainly points in that direction.

Figure from Peters, Jan, and Stefan Schaal. "Reinforcement learning of motor skills with policy gradients", *Neural networks*, 2008

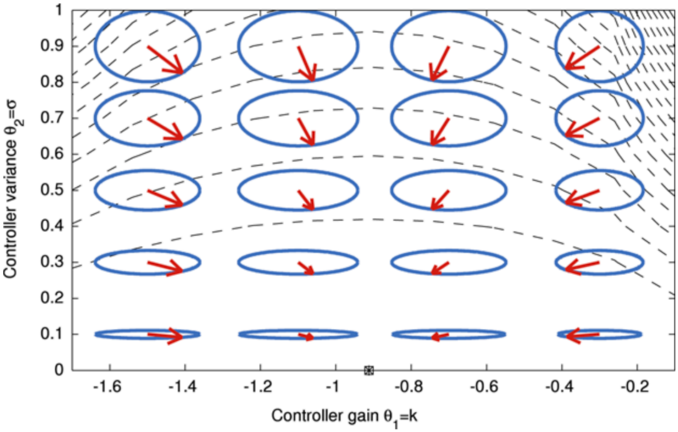
Example: Vanilla Policy Gradient for LQG



- Control policy: $u_t \sim \mathcal{N}(\theta_1 x_t, \theta_2)$, $\theta \leftarrow \theta + \alpha \widehat{\nabla V_{\pi_\theta}}$
- Decrease of exploration has a stronger immediate effect on the expected return. So, the gradient mainly points in that direction.
- This will quickly reaches the 'plateau of zero exploration' and results in a very slow convergence

Figure from Peters, Jan, and Stefan Schaal. "Reinforcement learning of motor skills with policy gradients", *Neural networks*, 2008

Example: Natural Policy Gradient for LQG



- Control policy: $u_t \sim \mathcal{N}(\theta_1 x_t, \theta_2)$, $\theta \leftarrow \theta + \alpha \widehat{F}(\theta)^{-1} \widehat{\nabla V_{\pi_\theta}}$

Figure from Peters, Jan, and Stefan Schaal. "Reinforcement learning of motor skills with policy gradients", *Neural networks*, 2008.